A comprehensive experimental study of thermal response mechanisms of a typical energetic material

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I. Introduction

The experimental research of this project was conducted in the Combustion Laboratory of Northwestern Polytechnical University. This laboratory has a high-pressure chemical shock tube experimental system and a low-pressure single-pulse cracking shock tube experimental system, which can obtain fuel ignition delay time and pyrolysis product components and concentrations. This project not only uses a shock tube, but also uses a gas chromatograph, so that a single shock tube experiment can simultaneously obtain the two parameters of ignition delay time and combustion product distribution and concentration, improve the experimental efficiency, provide richer parameters, and can be more In-depth verification of the kinetic mechanism of fuel combustion chemical reactions.



Figure 1. System installation diagram of high-pressure shock tube (left) and single-pulse shock tube (right)

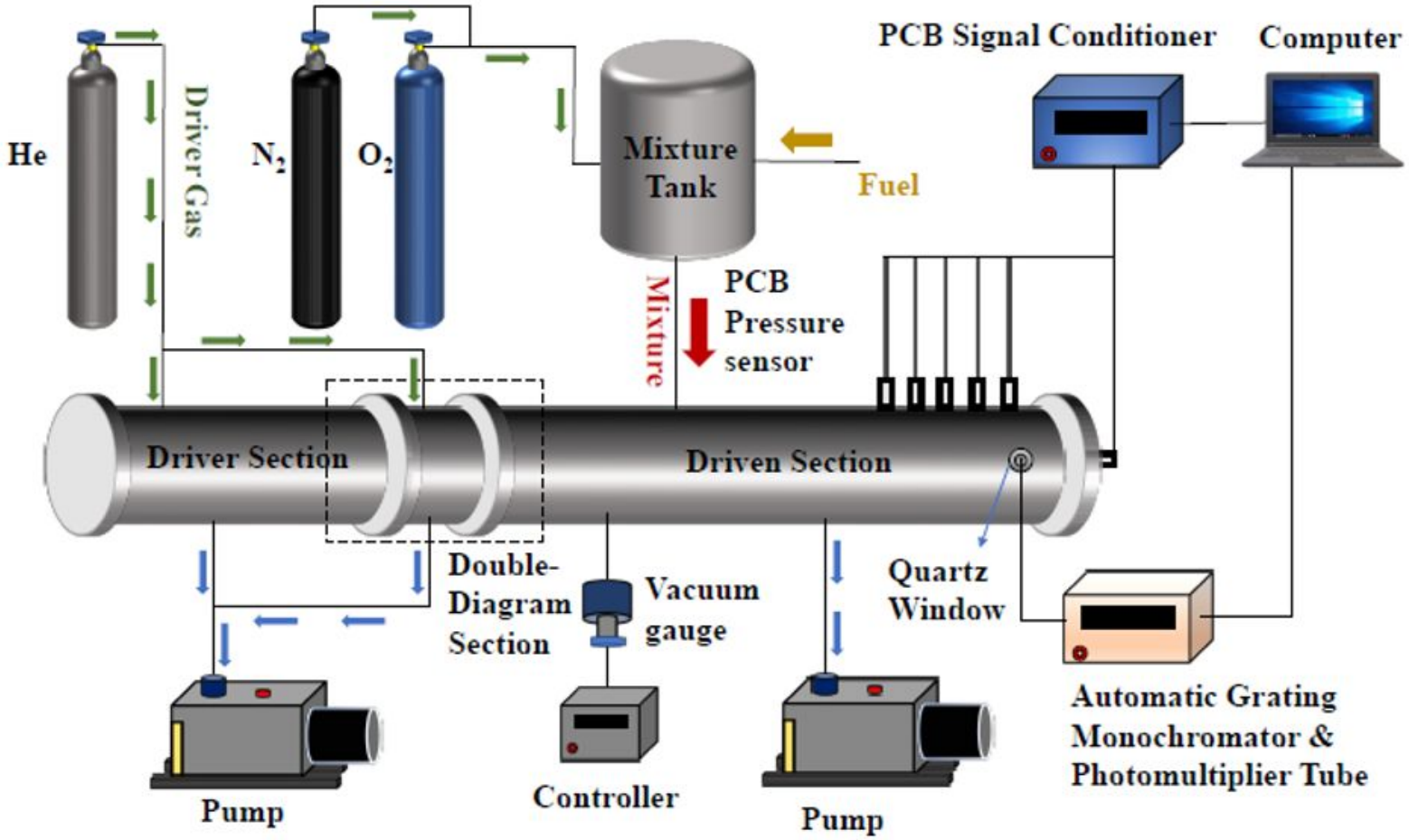


Figure 2. Schematic diagram of the high-pressure shock tube experiment system

II.

Theoretical Background

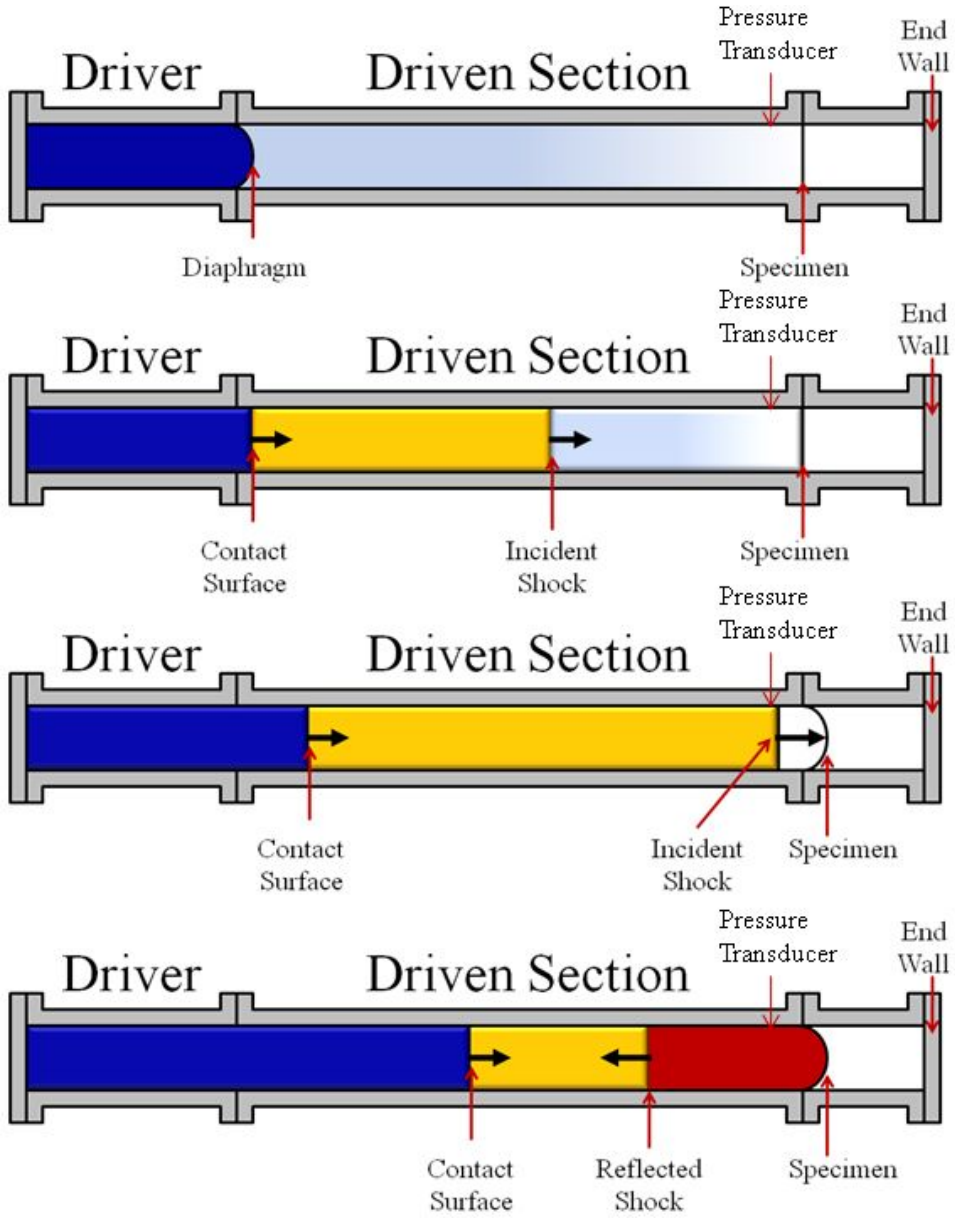
1- Shock Tube

Shock Tube is an experimental device used to generate high-temperature and high-pressure conditions over very short timescales, widely applied in combustion and energetic material research.

A shock tube typically consists of two main sections:

- Driver Section: Filled with high-pressure gas (e.g., Helium).
- Driven Section: Contains the test gas or sample at lower pressure.

The two sections are usually separated by a diaphragm, or two for better stability and repeatability. When the diaphragm ruptures, a shock wave propagates into the driven section. This sudden compression raises the temperature and pressure of the test gas almost instantaneously.

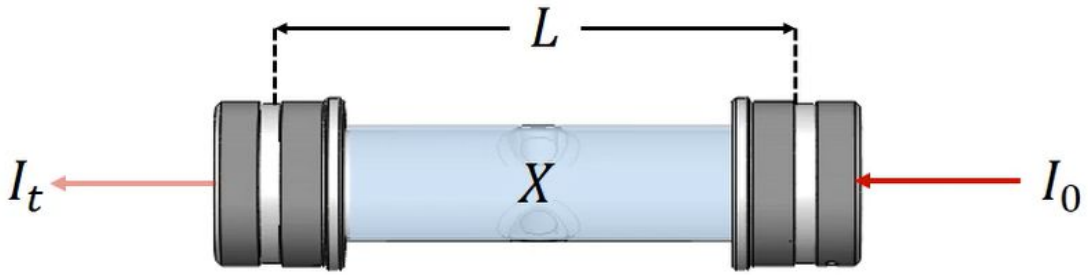


2- Beer's Law

Also known as the Beer-Lambert Law, is a fundamental principle used in spectroscopy to measure the concentration of a species in a sample. In combustion and shock tube experiments, this law is applied using laser absorption spectroscopy to analyze gases or fuels.

The basic principle of Beer's Law states that the absorbance of light passing through a medium is proportional to the concentration of the absorbing species and the path length. Mathematically, it is expressed as:

$$A=\epsilon LX$$



Where:

- A is the absorbance, representing how much light is absorbed (also denoted as α).
- is the molar absorption coefficient, which depends on the species and wavelength of light.
- L (or l) is the path length of light through the sample.
- X (or c) is the concentration of the absorbing species.

Since absorbance can be written in terms of light intensity, Beer's Law becomes:

$$A = -\ln ln\left(\frac{I_t}{I_0}\right)$$

When a beam of light (such as a laser) with an initial intensity I_0 passes through a gas sample, some of the light is absorbed by the species of interest in the gas. As a result, the transmitted intensity It exiting the sample is lower than I_0.

2- Beer's Law

In combustion research, Beer's Law is applied using laser absorption spectroscopy inside a shock tube. A shock tube generates a high-pressure, high-temperature environment where combustion or chemical reactions occur. The concentration of a specific species (like a fuel) in the reaction can be measured using Beer's Law.

Setup: A laser beam of a known wavelength is passed through a test section in the shock tube where the gas sample is located. This gas sample contains the species of interest whose concentration needs to be measured.

Initial and Transmitted Intensities: To ensure accuracy, the initial intensity I0 is measured when the sample cell is in a vacuum (no absorption). This eliminates errors due to reflections or window absorption. The transmitted intensity It is then measured when the gas sample is introduced. The ratio ItI0 indicates how much light was absorbed, from which we calculate the concentration X.

Revised Beer's Law for Combustion Applications: For practical systems, the molar absorption coefficient is replaced with the absorption cross-section (obtained from databases)

$$\epsilon = \sigma \times n = \sigma \times \frac{N}{V} = \sigma \frac{P}{RT}$$

The revised equation then becomes:

$$X = \frac{-\ln ln \left(\frac{I_t}{I_0}\right) RT}{P\sigma L}$$

Where:

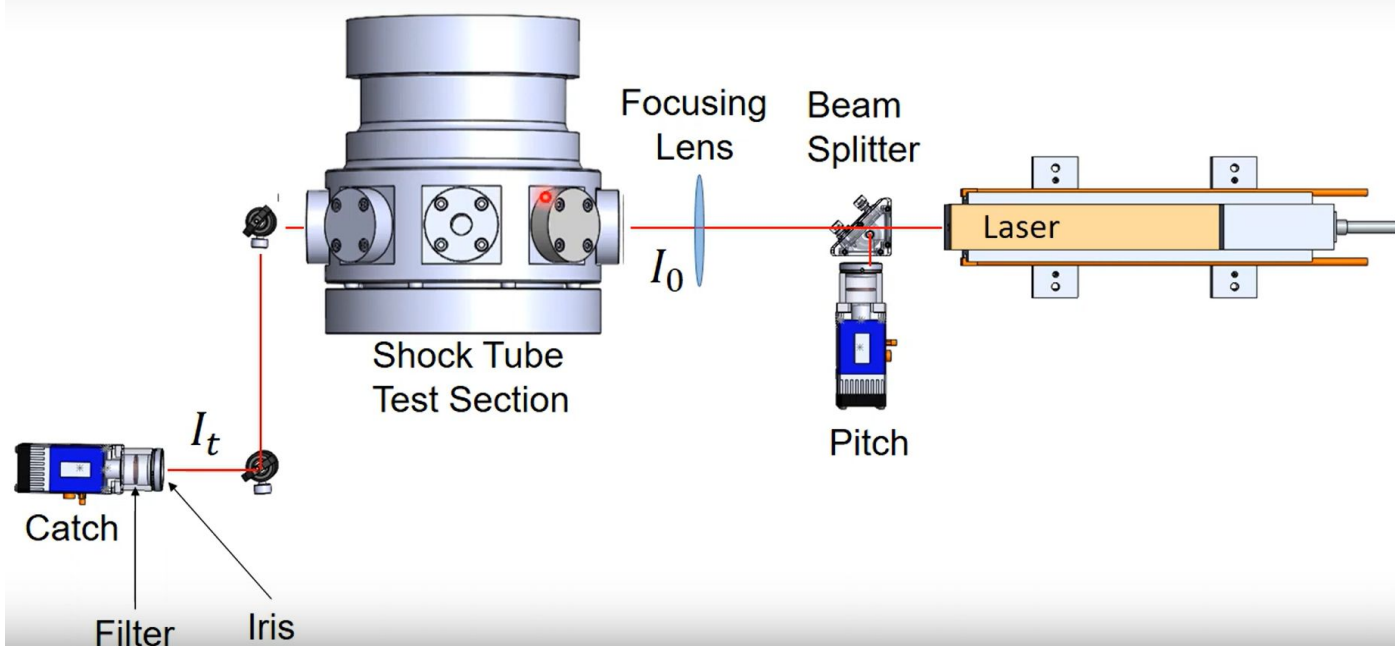
- R is the universal gas constant (8.3145 J/mol·K).
- T is the temperature (measured with a thermocouple).
- P is the pressure (measured with a transducer).
- σ is the absorption cross-section (dependent on species, temperature, and wavelength).

2- Beer's Law

Experimental setup:

A laser beam of a specific wavelength is passed through the test section containing a sample gas. The setup consists of a **Laser Source** which generates light of a known wavelength absorbed by the target species, **Beam Splitter** which divides the beam, sending part of it to a "pitch" detector for reference and the rest through the sample and also monitors the laser's output to correct for natural fluctuations. **Test Section** which contains the absorbing species, and the transmitted beam exiting the test section with reduced intensity It. Detectors (Catch and Pitch), where the **Catch detector** measures transmitted intensity It, and the **Pitch detector** monitors variations in the laser output to account for signal drift.

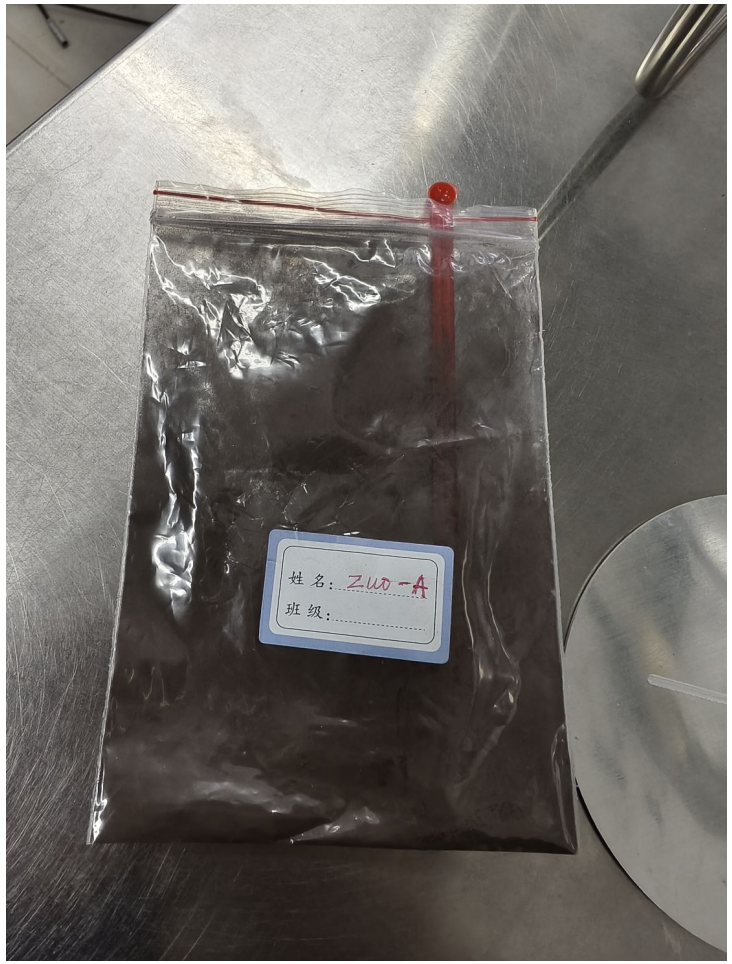
The setup does not fully eliminate errors caused by laser fluctuations over time, as It and I0 are measured sequentially. Techniques like beam steering correction, using focusing lenses, irises, and wavelength filters, help reduce these errors**:**

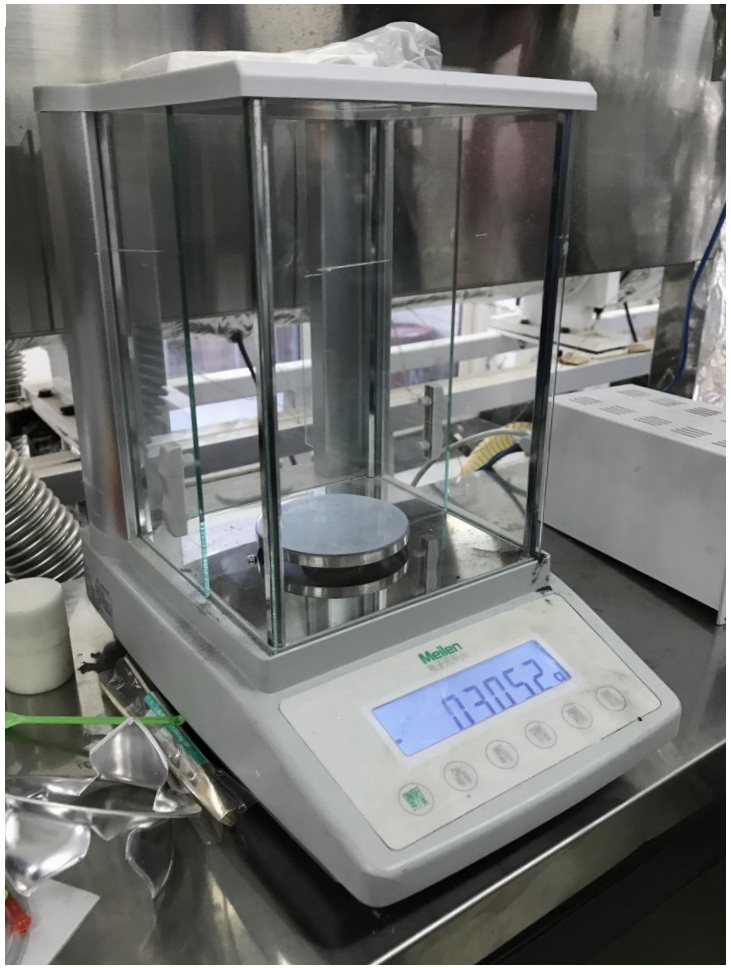


III.

Experimental Process

First, Accurately measure the mass of the dynamite powder using a precision balance to ensure consistency across all experiments.





Second, Operate the control system to set the pressure conditions in both the low-pressure and high-pressure sections.



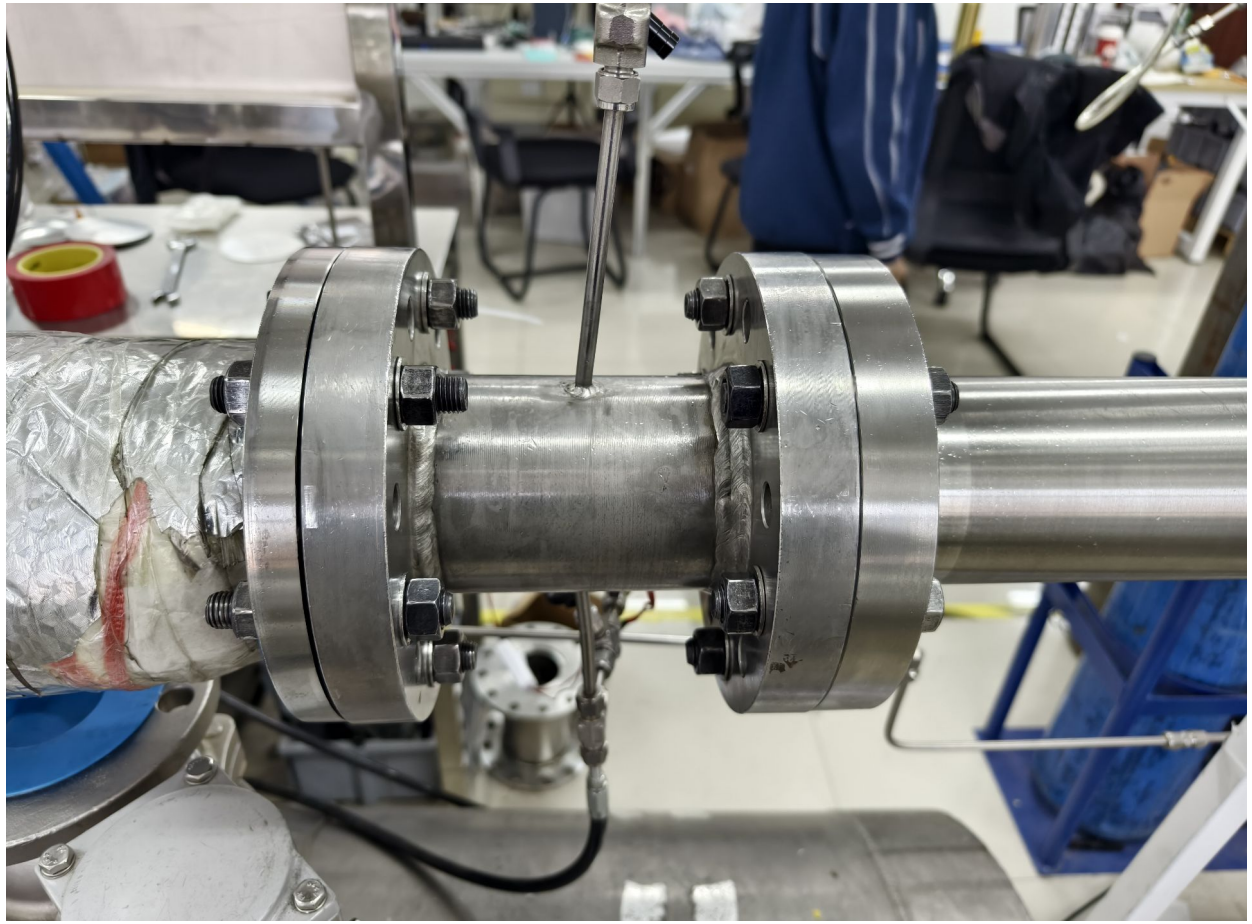
Third, Charge the gas reservoirs with Nitrogen and Helium, following the pre-determined pressure ratios derived from prior calculations.



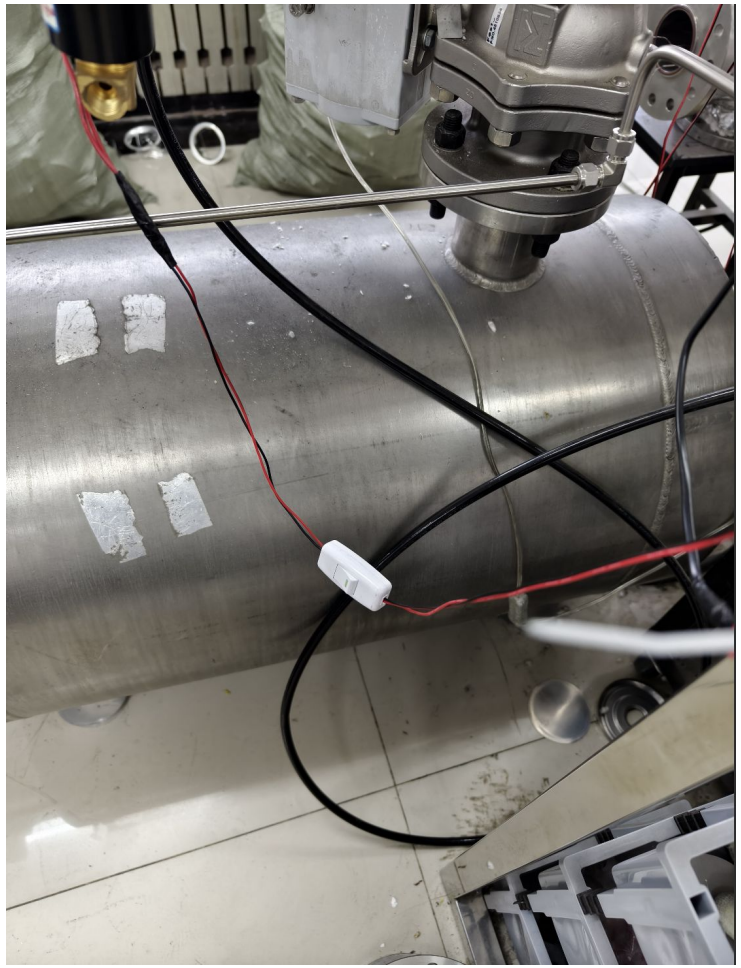


Fourth, Insert two membranes into the designated double diaphragm section of the shock tube. Ensure proper alignment for consistent rupture behavior.





Fifth, Start the experimental setup by activating the corresponding control button. The system will generate a controlled pressure wave upon membrane rupture.





Lastly, Release the residual gas from the shock tube by operating the exhaust control system. Safely dispose of the ruptured membranes and prepare the setup for subsequent trials.

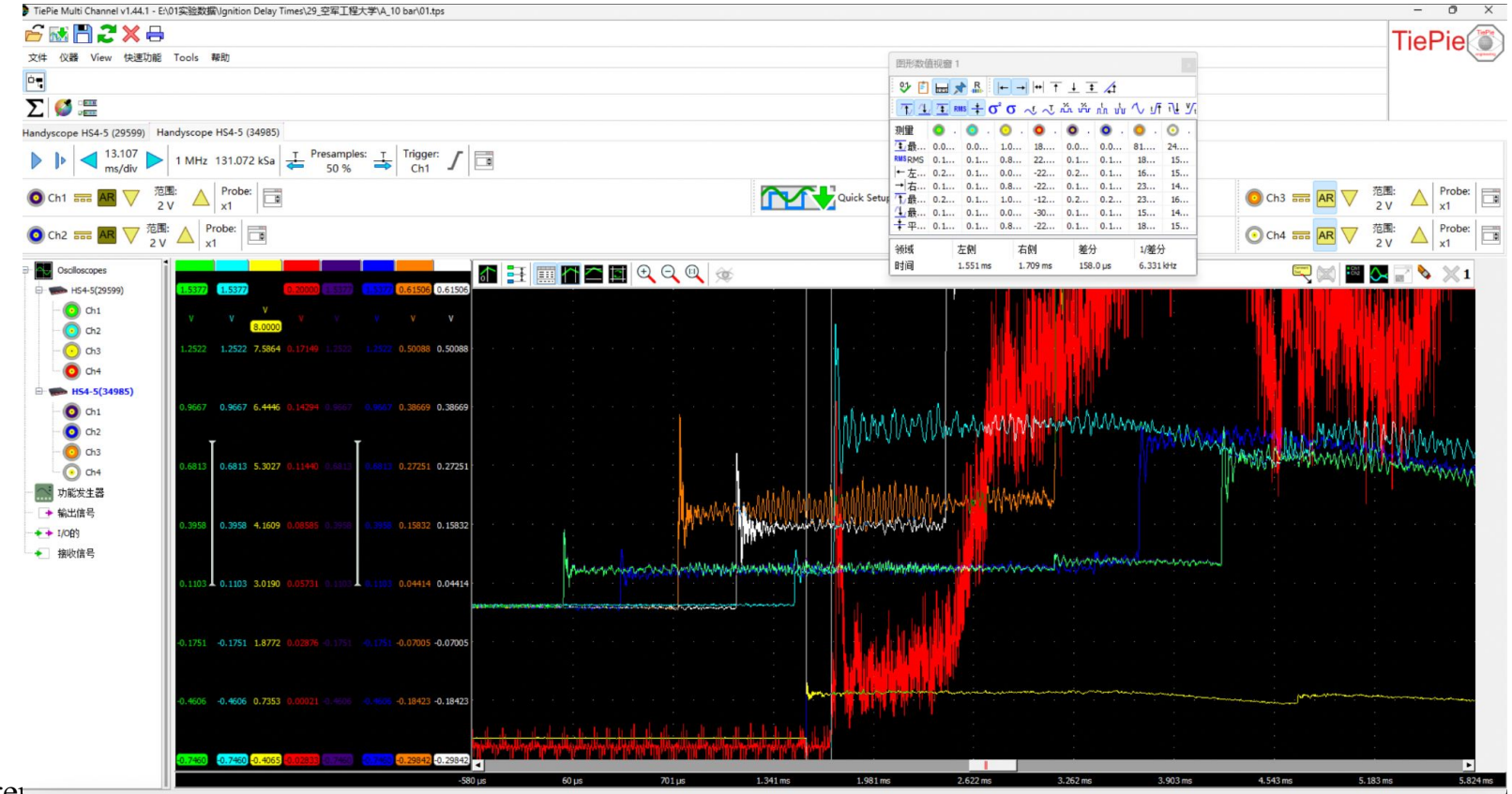


IV. Results and Discussion

The experimental results are analyzed through the recorded signals, represented in the diagrams.

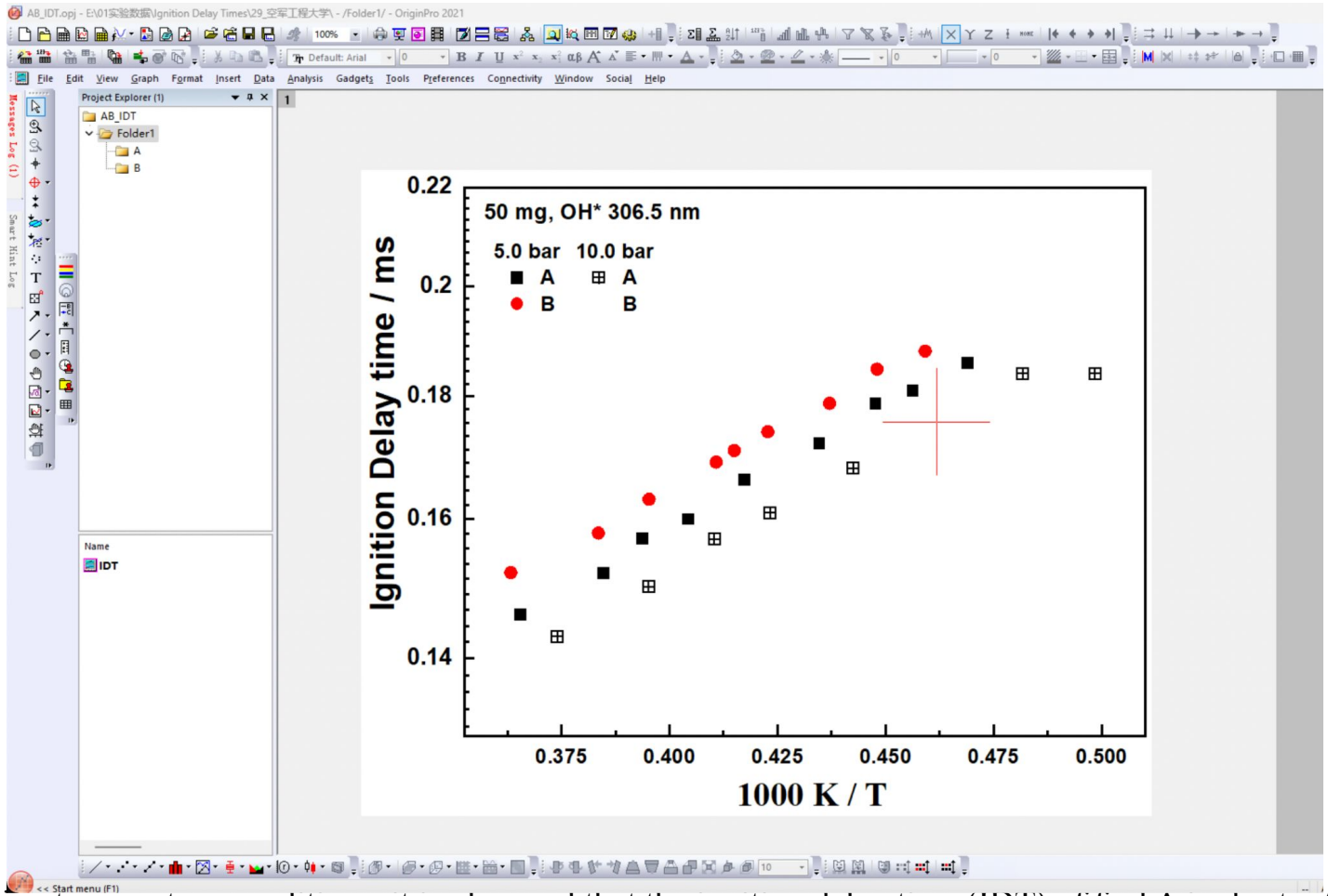
The red signal corresponds to the grating response, marking the precise moment of ignition occurrence.

The yellow signal indicates the initiation of the high-temperature and high-pressure region within the tube, signifying the onset of the reactive environment.



The temporal difference between the red and yellow signals represents the ignition delay time (IDT). This parameter quantifies the duration from the start of the experiment, where the reaction conditions are established, to the ignition of the fuel. The IDT is critical for evaluating the reactivity and thermal response characteristics of the energetic material under the specified conditions.

The red curve corresponds to the performance of fuel A, while the black curve represents fuel B. To analyze the data, a specific point is selected, for instance, 0.400. At this point, the temperature is 1000 K, and the reciprocal relationship is calculated as 1000 K/0.400=2500, indicating an operating temperature of 2500 K.



Under identical operating temperature conditions, it is observed that the ignition delay time (IDT) of fuel A is shorter than that of fuel B. This shorter IDT suggests that fuel A exhibits superior reactivity and thermal response compared to fuel B, making it more effective under the tested conditions.

Thank you for your attention!



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